
Continual Egocentric/Exocentric Machine Perception: From Procedural Understanding to Embodied Action

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SUMMARY

Current multimodal models struggle with long-horizon egocentric and exocentric understanding because they process continuous experience mainly through sampled context, short temporal windows, or autoregressive prediction. This leaves open a central question: *what context-relevant information should be retained, updated, and reused over time?* This proposal formulates Continual Egocentric/Exocentric Machine Perception as *Continual Procedural Learning*: The problem of constructing, updating, and using procedural state representations from continuous multimodal streams. The research will develop methods for procedural state estimation, benchmarks for long-horizon procedural reasoning, and downstream evaluations in mistake detection, worker guidance, collaborative task understanding, and embodied action. The expected outcome is a reusable research foundation for long-horizon procedural understanding, supported by open methods, benchmark definitions, and evaluation protocols. The work will be developed responsibly and aligns with research directions in multimodal assistants, long-video understanding, embodied AI, and reliable real-world deployment.

INTRODUCTION AND MOTIVATION

Advances in Machine Learning (ML), Computer Vision (CV), Self-Supervised Learning (SSL), model scaling, and multimodal foundation models have substantially improved perception, representation learning, and context-conditioned reasoning [1, 2, 3, 4, 5, 6, 7]. In video understanding, the field is moving from short-clip recognition toward long-horizon, multimodal reasoning over extended temporal context [8, 9, 10]. Egocentric perception extends this setting by adding embodiment and agency: the camera moves with the acting agent, hands and objects define task relevance, attention and intent matter, and the system may need to support future action. Ego4D [11] and Ego-Exo4D [12] formalize this shift by connecting first-person perception with episodic memory, hand-object interaction, forecasting, procedure understanding, and complementary exocentric context. Thus, procedural understanding requires temporal reasoning, semantic grounding, and embodied memory.

Scaling context length and autoregressive prediction can improve short-term recognition and generation, but that does not solve long-horizon procedural understanding. Real-world ego/exo data is noisy, partially observed, and dominated by irrelevant or redundant content. A useful system must decide what information should persist over time, segment continuous activity into meaningful events, track object states before and after actions, ground language in specific moments and objects, reason over procedural dependencies, and represent uncertainty when evidence is incomplete. Current VLMs and VLLMs often produce plausible semantic interpretations, but remain weak at maintaining persistent task state, detecting missing or incorrect steps, temporal grounding of outputs, and supporting real-time assistance or planning under bounded compute and memory. This presents a broader problem for perception and embodied AI: systems that act in the physical world require mechanisms for selectively encoding, updating, and retrieving task-relevant procedural state from multimodal experience. Without these mechanisms, downstream action cannot be reliably grounded, verified, or adapted.

RESEARCH PROBLEM AND CORE HYPOTHESIS

Ego/exo perception is inherently continual. A wearable assistant, a robot-mounted camera, or a stationary workspace camera observes procedures as continuous multimodal streams rather than as clean, curated samples. These streams include video, language, motion, gaze, interaction cues, object state changes, and environmental context, often distributed across long temporal horizons. This setting changes the formulation of Continual Learning (CL); the central issue is not only how to update a model across predefined classes or tasks, but how an embodied system should incrementally retain, compress, update, and retrieve task-relevant information from

ongoing experience. For procedural understanding and embodied action, this includes memory of objects, locations, actions, task progress, user or agent behavior, object states, and changes over time.

The central hypothesis is that real-world ego/exo and embodied perception requires *Continual Procedural Learning (CPL)*: the ability to incrementally construct, update, and query structured memory from continuous multimodal experience. In physical environments, an agent must infer what has changed, what matters, what should be retained, and how current observations relate to prior task context. This is motivated by human learning, where continuous experience is organized into meaningful events, linked to prior knowledge, and selectively retained for future behavior [13]. Humans use perception, action, prediction, memory, and feedback to maintain grounded state representations over time. Current foundation models encode rich semantic knowledge, but they do not reliably implement this cycle of structured encoding, selective retention, and experience-grounded updating. This research develops computational methods for constructing such procedural memory.

This framing connects human assistance, robotic perception, and foundation-model research by treating procedural state as something inferred from sparse multimodal evidence rather than dense labels or predefined task graphs. In a worker-assistance setting, procedural memory can support documentation, task guidance, and error detection by maintaining an interpretable estimate of task progress and relevant object states. In robotics, a robot-mounted camera provides an egocentric stream from the perspective of the acting agent, while stationary cameras provide exocentric context about the workspace, collaborators, and object layout. Foundation models are treated as powerful perceptual and semantic engines. VLMs, VLLMs, and VLAs can provide features, semantic priors, captions, candidate interpretations, and action proposals. CPL provides the mechanism for retaining, updating, and using process-relevant structure over time.

ESTABLISHED WORK AND PRELIMINARY EVIDENCE

Industrial scenarios provide a rigorous target domain for continual ego/exo perception because they combine visual variability, multimodal sensing, procedural structure, human-object interaction, and downstream relevance for assistance and robotic operation [14]. Our development of three open multimodal datasets provides the data foundation. InVar-100 [15] and MVIP [16] support research on object recognition under distribution shift and multi-view contexts. IndEgo [17] then moves from object-centric perception to long-horizon procedural understanding, providing synchronized egocentric and exocentric recordings with gaze, motion, narration, and annotations for industrial tasks such as assembly, disassembly, logistics, repair, and collaborative work. In parallel, work on Active Data Collection and Management (ADCM) [18] showed that even smaller SSL pretrained encoders can serve as feature-space oracles for selecting representative subsets from uncured data streams (image and video data). Together, these contributions establish the empirical basis for the proposed research.

Our evaluations show that CL methods often perform worse under realistic visual variation, varying increment lengths, and changing task order than under controlled academic benchmarks. Background context, visual noise, and distribution shift affect both the ability to learn new data and the retention of previously learned information [15]. The results also show that accuracy alone is insufficient for evaluating CL methods; computational footprint is necessary to assess practical efficacy. ADCM [18] improved exemplar memory selection and incremental accuracy over existing strategies. Additional experiments showed that saliency-aware semantic encoding, including regions of interest or gaze-informed saliency, can improve CL behavior, and that robust Self-Supervised Learning (SSL) representations are less vulnerable to forgetting than weaker supervised representations [19]. These findings support the proposed formulation of continual perception as a problem of selecting, structuring, and retaining task-relevant information under resource constraints.

Research on egocentric perception investigated how wearable devices (capturing RGB observations, narration, gaze, and interaction cues) can support industrial inspection and provide signals for attention, intent, and task relevance [20]. Building on this, IndEgo introduced benchmarks for procedural mistake detection, video question answering, egocentric data summarization, and collaborative task understanding in industrial scenarios [17]. These benchmarks test situated reasoning, temporal understanding, visual recognition, abductive reasoning, instruction following, and multi-person task modeling. Baseline evaluations with current VLMs and VLLMs show a substantial gap between existing multimodal models and the requirements of reliable egocentric assistants and embodied agents, especially for long-horizon reasoning, procedural mistake detection, collaboration, and ego/exo grounding. Follow-up work on AI-based worker guidance showed that structured task representations derived from expert demonstrations and narration can support procedural documentation and context-aware next-step guidance in assembly and disassembly tasks [21].

PLANNED WORK AND EVALUATION

Building on the established foundation, the proposed work moves from passive recognition toward systems that estimate procedural state, retain task-relevant information, and use this state for reasoning and action.

Methods: The core technical component of CPL is a procedural state representation and update method. Given continuous ego/exo streams, the system will identify task-relevant temporal units, abstract them into procedural events, update the current procedural state, and retrieve prior causal context when needed. The representation will encode task progress, objects and tools, object states, actions, agents, relations, temporal dependencies, uncertainty, and deviations. It will be learned from sparse multimodal cues such as video, gaze when available, motion, hand-object interaction, object state changes, occasional narration, and pretrained VLM/VLLM features, rather than dense step annotations or predefined task graphs. The goal is to maintain a compact and queryable state that supports long-horizon reasoning without storing or processing the raw stream exhaustively.

Evaluation: Building on existing ego/exo industrial data, the proposed benchmark will test whether models can maintain and use procedural state over time. Evaluation tasks will include task progress estimation, current-step recognition, next-step prediction, missing-step detection, order violation detection, procedural mistake detection, task summarization, long-horizon question answering, ego/exo grounding, and collaborative dependency modeling. Baselines will include raw-video prompting, short-window VLM/VLLM inference, retrieval-based video agents, flat text summaries, task-graph approaches, and the proposed procedural state methods. The key evaluation criterion is whether explicit procedural state improves reasoning under memory, compute, and data-efficiency constraints.

Validation: CPL will be evaluated in downstream assistance and embodied-action settings. For human-centered assistance, the system will be tested on real-time task guidance, procedural documentation, mistake detection, and next-step recommendation across diverse industrial scenarios. For embodied agents, procedural state will be used to condition robot policies or VLAs through subgoals, object-state transitions, task constraints, and expected next actions. The evaluation will test whether CPL improves long-horizon reasoning, robustness to distractors or changed object configurations, recovery from procedural deviations, and task success under bounded compute and memory. Near-term work will formalize the procedural state representation and update mechanism; mid-term work will evaluate CPL on long-horizon procedural reasoning and assistance tasks; longer-term work will validate CPL through real-time guidance and embodied execution experiments.

IMPACT, RESPONSIBLE DEPLOYMENT, AND FUTURE DIRECTION

The research follows an open-science strategy through publications, benchmark releases, baseline implementations, evaluation protocols, open-source code, and collaborations. These outputs will support broader work on continual multimodal perception, egocentric assistants, long-horizon procedural reasoning, and embodied AI. Industrial procedures provide a practical validation setting for CPL because procedural knowledge is economically valuable, but difficult to document, transfer, and reuse. TU Berlin provides the research environment, industrial collaboration (including subject matter expertise), digital infrastructure, and robotics expertise needed for a realistic evaluation. Responsible deployment is integral; human data is collected under informed consent and GDPR-compliant governance, assistance systems are designed to augment rather than replace human decision-making, human oversight is preserved in safety-critical settings, and risks such as covert monitoring are addressed through purpose limitation, data minimization, transparent data use, anonymization, and evaluation across users and application scenarios [22].

CONCLUSION

This proposal addresses a fundamental bottleneck in machine perception: long-horizon egocentric and exocentric understanding is often treated as a problem of larger context windows, denser sampling, or stronger foundation models, while the question of what task-relevant state should be retained and updated over time remains unspecified. The proposed research builds on current progress in pretraining, multimodal foundation models, and VLM/VLLM/VLA systems, but studies the complementary problem of Continual Procedural Learning: *how systems can construct, update, and use procedural state from continuous multimodal experience*. The expected outcome is a set of methods, benchmarks, and open tools for evaluating whether models can maintain procedural state over long horizons and use it for reasoning, mistake detection, worker guidance, and embodied action. By connecting continual perception, procedural state, and action, this research aims to establish a rigorous foundation for ML systems that actively process physical experience, and also retain, verify, and act on it.

REFERENCES

- [1] Alex Krizhevsky et al. “ImageNet Classification with Deep Convolutional Neural Networks”. In: *Advances in Neural Information Processing Systems*. Ed. by F. Pereira et al. Vol. 25. Curran Associates, Inc., 2012. URL: <https://proceedings.neurips.cc/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf>.
- [2] Alexey Dosovitskiy et al. “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale”. In: *International Conference on Learning Representations*. 2021. URL: <https://openreview.net/forum?id=YicbFdNTTy>.
- [3] Alexander Kirillov et al. “Segment Anything”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2023, pp. 4015–4026.
- [4] Mathilde Caron et al. “Emerging Properties in Self-Supervised Vision Transformers”. In: *Proceedings of the International Conference on Computer Vision (ICCV)*. 2021.
- [5] Jared Kaplan et al. “Scaling laws for neural language models”. In: *arXiv preprint arXiv:2001.08361* (2020).
- [6] Tom B. Brown et al. *Language Models are Few-Shot Learners*. 2020. DOI: [10.48550/ARXIV.2005.14165](https://doi.org/10.48550/ARXIV.2005.14165). URL: <https://arxiv.org/abs/2005.14165>.
- [7] Alec Radford et al. *Learning Transferable Visual Models From Natural Language Supervision*. 2021. arXiv: [2103.00020](https://arxiv.org/abs/2103.00020) [cs.CV]. URL: <https://arxiv.org/abs/2103.00020>.
- [8] Mahmoud Assran et al. “V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning”. In: *arXiv preprint arXiv:2506.09985* (2025).
- [9] Haotian Liu et al. “Visual Instruction Tuning”. In: *NeurIPS*. 2023.
- [10] Boqiang Zhang et al. *VideoLLaMA 3: Frontier Multimodal Foundation Models for Image and Video Understanding*. 2025. arXiv: [2501.13106](https://arxiv.org/abs/2501.13106) [cs.CV]. URL: <https://arxiv.org/abs/2501.13106>.
- [11] Kristen Grauman et al. “Ego4D: Around the World in 3,000 Hours of Egocentric Video”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. June 2022, pp. 18995–19012.
- [12] Kristen Grauman et al. *Ego-Exo4D: Understanding Skilled Human Activity from First- and Third-Person Perspectives*. 2024. arXiv: [2311.18259](https://arxiv.org/abs/2311.18259) [cs.CV].
- [13] Jeffrey M Zacks and Khen M Swallow. “Event segmentation”. In: *Current directions in psychological science* 16.2 (2007), pp. 80–84.
- [14] Chiara Plizzari et al. “An Outlook into the Future of Egocentric Vision”. In: *International Journal of Computer Vision* 132.11 (Nov. 2024), pp. 4880–4936. ISSN: 1573-1405. DOI: [10.1007/s11263-024-02095-7](https://doi.org/10.1007/s11263-024-02095-7). URL: <https://doi.org/10.1007/s11263-024-02095-7>.
- [15] Vivek Chavan et al. “Towards Realistic Evaluation of Industrial Continual Learning Scenarios with an Emphasis on Energy Consumption and Computational Footprint”. In: *Proceedings of the International Conference on Computer Vision (ICCV)*. 2023.
- [16] Paul Koch et al. *MVIP: A Dataset for Industrial Part Recognition*. Fraunhofer Fordatis. 2023. DOI: <https://dx.doi.org/10.24406/fordatis/300>. URL: <https://fordatis.fraunhofer.de/handle/fordatis/358?mode=full&locale=en>.
- [17] Vivek Chavan et al. “IndEgo: A Dataset of Industrial Scenarios and Collaborative Work for Egocentric Assistants”. In: *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*. 2025. URL: <https://neurips.cc/virtual/2025/poster/121501>.

- [18] Vivek Chavan et al. “Active Data Collection and Management for Real-World Continual Learning via Pretrained Oracle”. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*. June 2024, pp. 4085–4096.
- [19] Vivek Chavan et al. “A System 1 and System 2 Perspective on Continual Learning for Practical Implementation”. In: *Computer Vision – ECCV 2024 Workshops*. Ed. by Alessio Del Bue et al. Cham: Springer Nature Switzerland, 2025, pp. 320–332. ISBN: 978-3-031-91578-9.
- [20] Vivek Chavan et al. “On the Application of Egocentric Computer Vision to Industrial Inspection”. In: *European Conference on Computer Vision (ECCV) Workshops*. Oct. 2024.
- [21] Vivek Chavan and Jörg Krüger. “AI-based worker guidance in assembly and disassembly operations using multimodal ego/exo-centric data capture and structured task knowledge”. In: *CIRP Annals* (2026).
- [22] Vivek Chavan et al. *Feeling Machines: Ethics, Culture, and the Rise of Emotional AI*. 2025. arXiv: [2506.12437](https://arxiv.org/abs/2506.12437) [cs.HC]. URL: <https://arxiv.org/abs/2506.12437>.